

STARTUP ZONE

BLDC Fans Breaking The Efficiency Ceiling

NIDHI ARORA



Expanded view of motor with various parts and remote

With the vision to develop smart home appliances that not only solve India's energy crisis but also bring in a wonderful experience and comfort to end customers, Manoj Meena and Sibabrata Das founded Atomberg Technologies to manufacture highly energy-efficient ceiling fan 'Gorilla.'

Traditional ceiling fans running on the old induction motor technology are highly inefficient. No innovation has been done in the fan industry for decades.

Meena and Das saw this as a big opportunity as well as responsibility.

Their ceiling fans run on BLDC motor technology and offer the following advantages:

1. Only 28W power consumption as against 75W by ordinary fans, enabling 65 per cent saving with similar output in terms of air delivery and RPM
2. Speed control with a remote
3. Timer and sleep modes integrated
4. Three times longer running time on inverter
5. Noiseless operation
6. No heating of the motor, which results in its long life

7. Consistent output even with fluctuating input voltage

8. Internet of Things (IoT) framework to connect appliances or other applications

9. Break-even period of less than one year—which is better than LEDs

Atomberg has sold more than 150,000 units of these fans since their launch two years ago. Sixty per cent of the shipments are made to B2B channels. To scale up, the company is now focusing on B2C segment. A pilot to understand retail has already started in Mumbai. Once the model is cracked, the company will replicate the same in tier-1 cities followed by tier-2 cities. It is also strengthening its online channel and expects 20-25 per cent online sales in the coming time.

After graduating from IIT Bombay in 2011, Meena and Das bootstrapped for initial three years by undertaking data logging and process control projects for IITs and other R&D institutes. In this duration, they gained expertise in electronics hardware and motor development. Once motor prototype was developed, their biggest challenge was to set up a manufacturing facility and build a network of reliable vendors. Yet another challenge was to educate various customers about the BLDC technology and its energy-saving benefits.

BLDC stands for Brushless Direct Current. As their name suggests, BLDC motors have no mechanical brush for commutation of windings. Commutation is deployed with the help of smart electronics, which senses the position of magnet rotor with respect to stator and controls the motor-driving switches. Winding's magnetic field reacts with the field of permanent magnets on the rotor to develop the required torque.

Gorilla motor design is optimised mechanically and electronically to minimise losses due to heating. This